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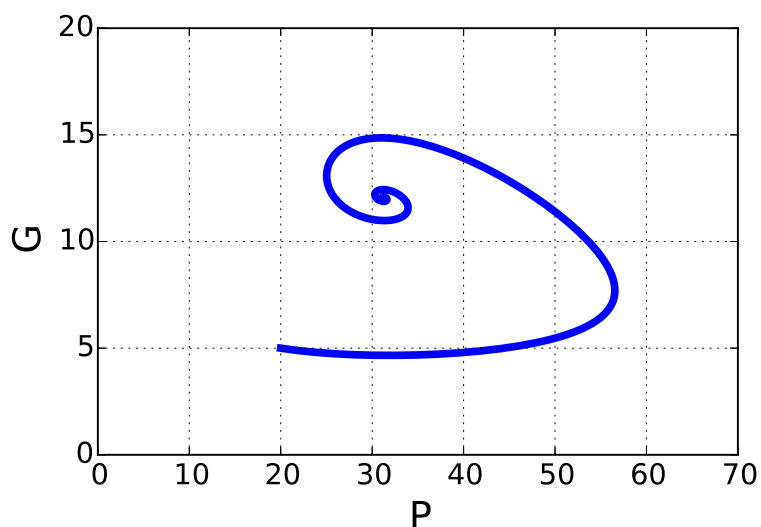
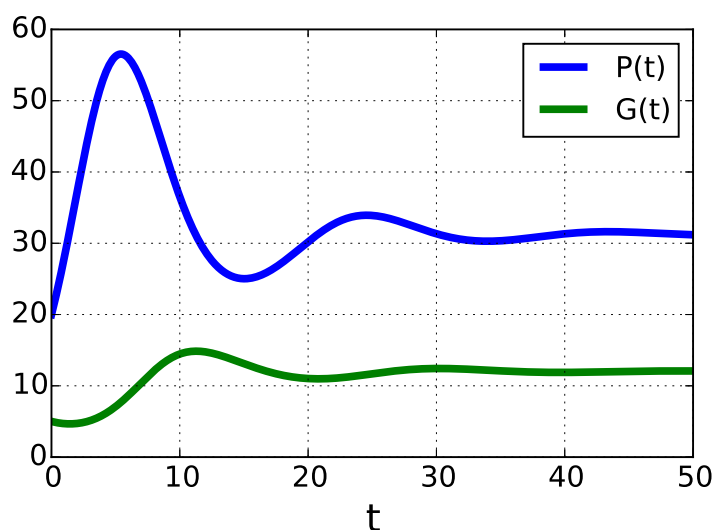
Validation

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The problem was

A client of the fish farm wants an aquarium with 20 rainbowfish, and 5 gouramis. Is it possible to have both fish in the same aquarium without one of them going extinct?

After constructing a mathematical model (for which the numbers are admittedly soso), graphs of the resulting solutions for both populations are



So the number of rainbowfish will grow fast from the initial 20 to about 55 in the first week. Because of the plentyfull rainbowfish, the number of gouramis will triple in a week and a half. The gouramis will eat more and more of the rainbowfish, so the number of rainbowfish will decrease again to about 25 after two weeks. With fewer rainbowfish, the number of gouramis decreases again, giving the rainbowfish population the chance to grow again. And so on, and so on. The increases and decreases of both populations become smaller in time, and after a month and a half, both populations stabilize to approximately 31 rainbowfish and 12 gouramis.

Exercise A

1/1 point (graded)

Which of these words are applicable to the solutions $P(t)$ and $G(t)$ shown above?

☐ monitonically increasing (always increasing or constant)

☐ monitonically decreasing (always decreasing or constant)

☒ bounded

☐ unbounded

☐ periodic (Is there is a period T such that $P(t + T) = P(t)$ and $G(t + T) = G(t)$ for all time t ?)

☒ oscillating

☐ non-oscillating

☐ undamped

☒ damped



Answer

Correct: Perfect, you got them all!

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You have used 1 of 3 attempts

The functions for the population sizes $P(t)$ and $G(t)$ are oscillating and damped. These solutions can be approximated by equations of the form

$$\begin{aligned}P(t) &\approx A + B e^{-\mu t} \sin(\omega(t - \phi)), \\G(t) &\approx C + D e^{-\mu t} \sin(\omega(t - \psi)),\end{aligned}$$

where the constants are named as follows:

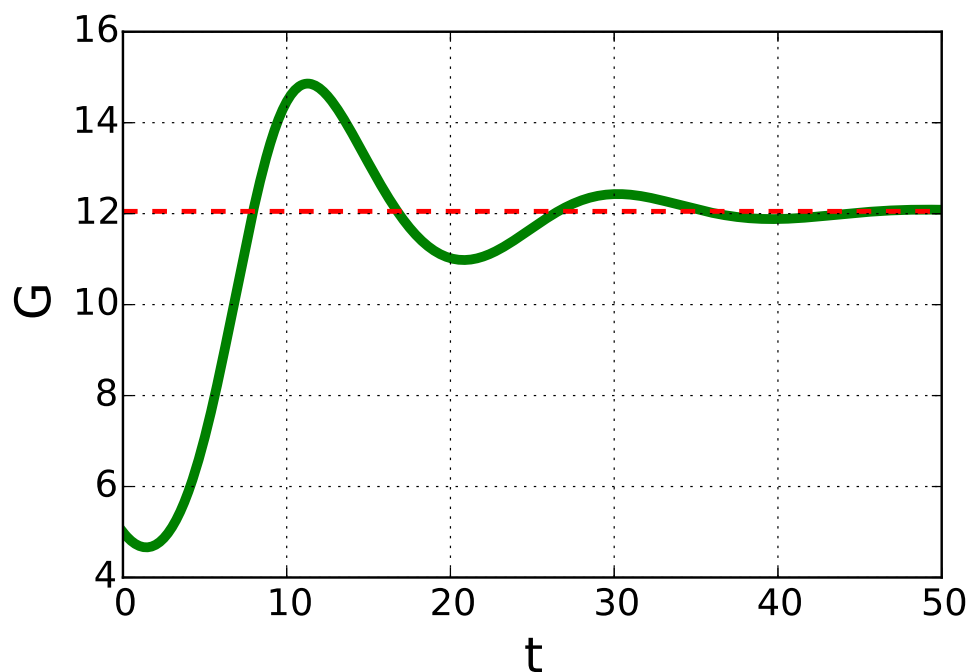
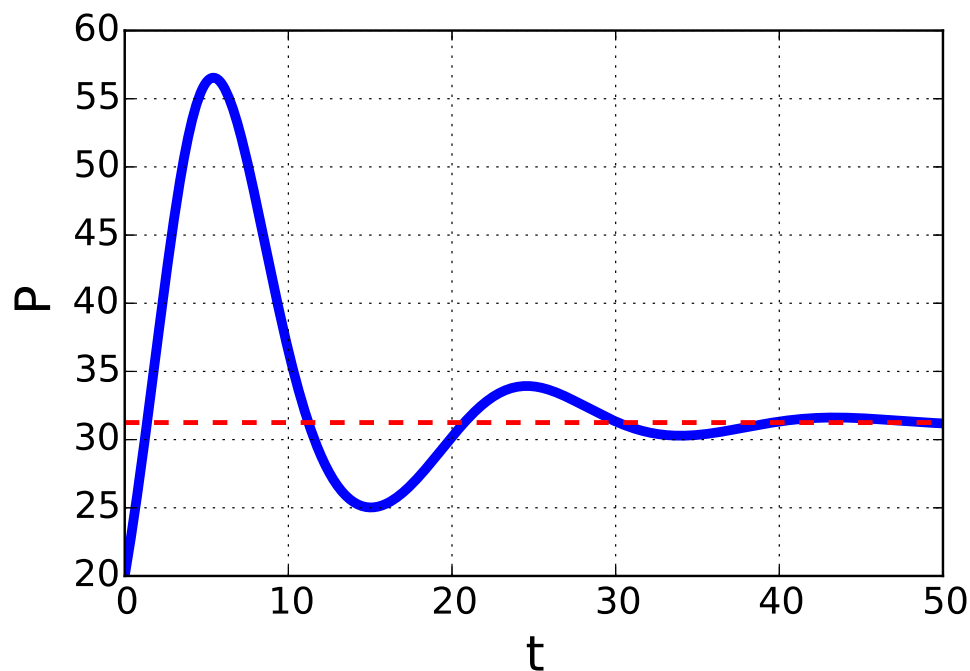
- *Vertical shifts* or mean values A and C , with $[A] = [C] = \text{fish}$;
- *Amplitudes* B and D , with $[B] = [D] = \text{fish}$;
- A *damping constant* μ , with $[\mu] = \text{day}^{-1}$;
- *Phase shifts* ϕ and ψ , with $[\phi] = [\psi] = \text{day}$;
- *Angular frequency* ω , with $[\omega] = \text{radians per day}$;

Also three other properties of the solutions can be derived from these formulas:

- The *period* $T = \frac{2\pi}{\omega}$, with $[T] = \text{day}$;
- The *frequency* $f = \frac{\omega}{2\pi} = \frac{1}{T}$, with $[f] = \text{day}^{-1}$;
- The *phase difference* $\psi - \phi$ in days.

These formulas only approximate the solutions of our system of differential equations for P and G (they are solutions of the linearised system of differential equations). Thus, it is not useful to try to fit all these constants very precisely for our solutions for P and G . However, the formulas do give us words to describe the cycles the populations go through.

In the next exercises you are going to estimate the average value of the populations A and C , the period of the cycles T , and the phase difference $\psi - \phi$ between the two populations. Here are again the graphs:



Exercise B

1/1 point (graded)

The vertical shift constants are the values around which the solutions oscillate, and the population sizes after a long time. A is this level for $P(t)$, C for $G(t)$. Give estimates for A and C (in numbers of fish).

An estimate for A is:

✓ Answer: 31.0

32

An estimate for C is:

✓ Answer: 12.0

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i Answers are displayed within the problem

Exercise C

1/1 point (graded)

The populations go through a cycle that takes T days. The period can be estimated by measuring the time between to peaks following each other in the graph. The angular frequency ω can then be derived as $\omega = \frac{2\pi}{T}$. Give estimates for T (in days) and ω (in radians per day).

An estimate of T is:

✓ Answer: 19.25

An estimate of ω (with one or two decimals) is:

✓ Answer: 0.33

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Exercise D

1/1 point (graded)

Both the rainbowfish and the gourami populations follow a cyclic movement of T days: growth to a peak, then decrease again to a low, increase again, etc. The cycle of the gourami lags some days behind the one of the rainbow fish. How much is the cycle of the gouramis behind?

Estimate the difference in phases: $\psi - \phi$, by comparing for the two species the different peak days (and the low days).

An estimate of $\psi - \phi$ (in days) is:

✓ Answer: 6

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise E

0.6666666666666666/1 point (graded)

After a month and a half or so, the populations have stabilised to about $\backslash(31\backslash)$ rainbowfish and $\backslash(12\backslash)$ gouramis. Now we can see how the constant $\backslash(\alpha\backslash)$ in the model affects our virtual fish.

For $\backslash(t\rightarrow\infty\backslash)$, how many rainbowfish does a gourami eat each day? Give your answer in two decimals.

✗ Answer: 1.24

$\backslash(\backslash)$

For $\backslash(t\rightarrow\infty\backslash)$, how many rainbowfish are eaten each day by the gouramis? Give your answer in whole rainbowfish.

✓ Answer: 15.0

$\backslash(\backslash)$

For $\backslash(t\rightarrow\infty\backslash)$, how many rainbowfish hatch every day? (Assume that the rainbowfish only die when they are eaten by a gourami.) Give your answer in whole rainbowfish.

✓ Answer: 15.0

$\backslash(\backslash)$

You have used 3 of 3 attempts

i Answers are displayed within the problem

Next, a page for questions you might have about this section.

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 English ▼

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